

Optimization Techniques

Lab n°8 - Simulated annealing

Exercise 1 In this lab, the aim is to solve the Traveling Salesman's problem by means of simulated annealing.

The file `USA_cap.txt` contains the coordinates (X, Y) of 48 state capitals of the USA. These will be the city the salesman must visit.

Mind that:

1. Simulated annealing algorithms require a **huge** number of steps ($\gg 1000 - 10000$)
2. It may be convenient to start with high temperature ($T_0 \sim 1000$).
3. If you choose exponential cooling schedule, one choice could be $\alpha = \tau^{1/N}$, where τ is the reduction you want to achieve (i.e., your final temperature will be $T_0\tau$) and N the number of steps.

It could be nice to keep track of the computed length and plot it.

The minimal tour has length 33523.

Hints

- To write the algorithm, you could start with a simplified version of the algorithm, with 5 cities and their mutual distance given by the table

	A	B	C	D	E
A	0.0	3.0	4.0	2.0	7.0
B	3.0	0.0	4.0	6.0	3.0
C	4.0	4.0	0.0	5.0	8.0
D	2.0	6.0	5.0	0.0	6.0
E	7.0	3.0	8.0	6.0	0.0

The minimal tour has length 19.

- The following function computes the total distance of a trip that visits the cities in the order given by `S`. `A` is the matrix containing the mutual distances between cities.

```
1: function out=dist(S,A)
2: nc=max(size(S));
3: out=0;
4: for l=2:nc
5: out2=A(S(l-1),S(l))^2;
6: out=out+sqrt(out2);
```

- Remember that you want to come back to city of departure. For this, you must append, to the end of the random generated list of cities, the first city.